

Print appropriate debug information

Most of the professional projects have macros like TRACE or DEBUG_PRINT defined globally. They typically provide multiple debug levels like DEBUG_CRITICAL, DEBUG_MEDIUM, and DEBUG_ALL. At compile time, one could choose to have the binary compiled with the appropriate debug level.

A small trick could be:

```
#define TRACE printf("\nTRACE: [%s]<%s>", \
    _FILE_, _FUNCTION_)printf
```

Now `TRACE("Value is %d", val);` simply expands to:

```
printf("\nTRACE: [%s]<%s>", \
    _FILE_, _FUNCTION_)printf("Value is %d", val);
```

In addition to function and file information, one can choose to add line numbers and timestamps into the TRACE prints, depending upon the project requirements.

Writing comments

Lastly, a very important point is to write precise comments. The prime motive behind writing comments is to make it easier for others to understand the code.

Some projects actually produce the API documentation from the code itself using tools like 'doxygen', and hence the commenting also needs to have specific formats.

In general, authors should keep in mind the target audience/maintainers when they write the comments, and hence the comments should focus more on the core algorithm/central logic than the language constructs. It is also very important to write the comments about any deviation/hacks made during implementation.

Apart from the source files, even the makefiles, build scripts, etc. that are used for a project, should also have sufficient comments and README files available in them.

Writing a proficient C code is not a skill to be learnt overnight but something we learn when we code and read the code written by the gurus. The source code of the Linux kernel is one such project where programmers get to learn a lot about smart coding practices. 

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